

Utkarsh Singh Thakur

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Summary

Recent 2023 engineering graduate in **CSE with cyber security specialization**, seeking to get a head-start in the field of software development and cyber security. With a primary aim to leverage my technical knowledge and passion in organization's advancement and integrity while developing my skills and expertise further.

Experience

Cognizant Technology Solutions	January 2024 - June 2024
CSD Intern	Bengaluru
- Gained in-depth knowledge of Genesys Cloud platform architecture, features, call flows and best practices for optimal utilization. Developed strong problem-solving and troubleshooting skills while addressing integration challenges. - Designed and implemented efficient inbound call flows that optimized customer experience by streamlining call routing, reducing wait times, and ensuring callers reach the most appropriate agent for their needs. This involved leveraging features like interactive voice response (IVR) systems, skills-based routing, and call menus to automate initial call handling and efficiently direct customers to the correct department or agent.	

Skills

Python	MySQL	Cryptography
OOPS	C++	NumPy
Genesys Cloud	Cyber Security	

Education

SRM Institute of Science and Technology	2019 - 2023
Computer Science with specialization in Cyber Security	Bachelor of Technology
CGPA - 9.20	
Sainik School Ambikapur, Chhattisgarh	2016
CBSE	10th Standard
Sainik School Ambikapur, Chhattisgarh	2018
CBSE	12th Standard (PCMB)

Publications

Cost Minimization Statistical Method for Electricity Demand Forecasting using Multi Objective Learning Model	12/05/2023
The Seybold	
🔗 https://seyboldreport.org/article_overview?id=MDUyMDIzMDkxMjA2NDIOMzMx	

Projects

Cost Minimization Statistical Method for Electricity Demand Forecasting using Multi Objective Learning Model

- Developed and implemented a multi-objective learning model to accurately forecast electricity demand while minimizing computational costs.
- Combined Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks to optimize model performance and capture complex temporal dependencies in energy consumption data.
- Conducted rigorous data preprocessing to enhance data quality and improve model accuracy. Achieved superior forecasting results compared to state-of-the-art methods. Demonstrated it by lower Mean Absolute Percentage Error (MAPE) and Root Mean Squared Error (RMSE) metrics on the New York Independent System Operator (NYISO) dataset.

Data Security in Cloud using AES algorithm

- Conducted in-depth analysis of symmetric encryption algorithms to determine optimal data protection strategy. Selected AES as the most secure and efficient encryption method for cloud-based data.
- Justified AES selection through comparative analysis of various symmetric algorithms based on key size, block size, and resistance to attacks. Explored real-world applications of AES encryption to demonstrate its practical implementation in securing sensitive data.

Cloud Security Using Hash Algorithm

- Selected and implemented hashing, specifically SHA256, as the optimal cryptographic algorithm to protect against data tampering and unauthorized access. and demonstrated how hashing is used for security on cloud.
- Conducted a comprehensive evaluation of various encryption techniques to safeguard data integrity within the cloud environment. Justified the choice of SHA256 through comparative analysis of its superior collision resistance, avalanche effect, and overall security properties compared to other encryption algorithms.

Volunteering

Chhattisgarh Medical Association	2019, 2023
Blood Donation	

Interests

Fitness	Gaming
Travelling	AI/ML